

Enes Senel

Senior Scientist · Computational Biology

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SUMMARY

Senior Scientist with a decade of experience leading computational biology, single-cell multi-omics, and machine-learning programs across pharma, academia, and medical robotics. Currently leading multi-omics strategy in oncology and immunology at Johnson & Johnson Innovative Medicine. Co-author of widely used open-source tools (novoSpaRc, Optocoder), co-inventor on a US patent, and published in Nature Protocols and NAR Genomics & Bioinformatics. Strengths: technical leadership, reproducible production pipelines, cross-functional translation from data to therapeutic decisions, and mentoring scientists.

CORE SKILLS

Leadership: Technical leadership, scientific strategy, mentorship, stakeholder management, cross-functional collaboration

Computational Biology: Single-cell RNA-seq, CITE-seq, spatial transcriptomics, multi-omics integration, cell-cell interactions, biomarker discovery

Machine Learning: Deep learning, optimal transport, protein language models (ProtBERT), domain adaptation, computer vision

Engineering: Python, R, C++, CUDA, PyTorch, TensorFlow, Nextflow, scanpy, SQL, Docker, reproducible pipelines

Domain: Oncology, immunology, cell therapy, CMC analytics, pharma QC, medical robotics, neuroscience

PROFESSIONAL EXPERIENCE

Senior Scientist, Computational Biology

Johnson & Johnson Innovative Medicine · Berlin, Germany · Aug 2025 — Present

Technical lead for single-cell multi-omics in oncology and immunology — shaping computational strategy from discovery through product development.

- Lead end-to-end single-cell multi-omics programs (transcriptomics, proteomics, spatial) that directly inform target selection, mechanism-of-action narratives, and translational strategy for oncology and immunology assets.
- Set analytical direction across parallel projects — defining standards, reviewing methodology, and aligning computational deliverables with clinical and discovery stakeholders.
- Mentor junior scientists and manage external consultants; grow team capability through code review, design discussions, and structured onboarding.
- Partner with wet-lab, clinical, and CMC teams to translate computational insights into experimental designs and go/no-go decisions on assets in development.

Skills: Team leadership, Scientific strategy, Single-cell multi-omics, Oncology & immunology, Stakeholder management, Mentorship

Data Scientist, Computational Biology

Johnson & Johnson Innovative Medicine · Berlin, Germany · Oct 2023 — Jul 2025

Built the team's production analysis infrastructure and led single-cell projects across clinical and manufacturing programs.

- Architected and shipped automated high-throughput Nextflow/Python pipelines for scRNA-seq, CITE-seq, and multi-modal analyses — turning week-long manual workflows into reproducible runs.
- Led single-cell analyses underpinning biomarker discovery for a clinical-stage cell therapy and process characterization for CMC, contributing directly to regulatory-grade documentation.
- Established team conventions for reproducibility (containerization, version-controlled pipelines, QC standards) adopted as the default for new projects.
- Acted as go-to technical reviewer, raising the bar on rigor and reproducibility across the group.

Skills: Pipeline architecture, Nextflow, Python, scRNA-seq / CITE-seq, Reproducibility, Cell therapy biomarkers, CMC analytics

Doctoral Researcher — Computational Biology

Max Delbrück Center for Molecular Medicine · Berlin, Germany · Oct 2019 — Sep 2023

PhD developing and scaling open-source methods for spatial transcriptomics, adopted by labs and industry worldwide.

- Designed and led Optocoder, an ML pipeline for decoding barcoded transcripts from imaging-based spatial transcriptomics — published in NAR Genomics & Bioinformatics and adopted by external labs.
- Re-architected novoSpaRc (optimal-transport reconstruction of tissue architecture from scRNA-seq) for scalability, unlocking application to whole-tissue datasets; co-authored Nature Protocols paper.
- Co-inventor on a US patent for 3D spatial gene-expression reconstruction.
- Owned the full research lifecycle: problem framing, method development, benchmarking, open-source release, documentation, and supporting external users.

Skills: Method development, Spatial transcriptomics, Optimal transport, Open-source ownership, Scientific writing, Patents

Visiting Scientist

The Hebrew University of Jerusalem · Rehovot, Israel · Feb 2023 — Apr 2023

Independent collaboration applying protein LLMs to agricultural biotech discovery.

- Led an independent short-term project applying protein language models (ProtBERT) to discover novel anti-insecticidal proteins from raw sequence data — combining transfer learning with biological priors to guide downstream experimental validation.

Skills: Protein language models, Transfer learning, Cross-disciplinary collaboration

Machine Learning Research Engineer

Coriolis Pharma GmbH · Munich, Germany · Apr 2019 — Sep 2019

Delivered a production-ready deep-learning system for pharmaceutical QC.

- Developed and shipped deep-learning models for automated particle detection and classification from flow-microscopy images used in biopharmaceutical QC.
- Built an end-to-end Python/PyTorch/TensorFlow package — handed off to internal scientists as a self-serve tool.

Skills: Deep learning, Computer vision, PyTorch / TensorFlow, Pharma QC, Productization

Research Assistant — Computational Neuroscience

Max Planck Institute for Brain Research · Frankfurt am Main, Germany · May 2018 — Sep 2019

Multi-omics integration for activity-dependent neuroscience.

- Applied supervised ML to investigate activity-dependent changes in the neuronal proteome.
- Designed a multimodal integration and domain-adaptation pipeline bridging proteomic and transcriptomic datasets — foundational experience in cross-modality reasoning later applied to industry.

Skills: Multi-omics integration, Domain adaptation, Supervised ML

Undergraduate Research Assistant — Medical Robotics

Ozyegin University Robotics Lab · Istanbul, Turkey · 2014 — 2016

Real-time perception for an image-guided biopsy robot.

- Developed real-time needle-tip localization and tracking from ultrasound imaging, enabling autonomous robotic control for image-guided biopsy.
- Built a C++/CUDA software interface coupling real-time image analysis with robot control — first exposure to shipping latency-critical scientific software.

Skills: Real-time computer vision, C++ / CUDA, Medical robotics

EDUCATION

Ph.D. in Computational Biology

Humboldt University of Berlin · Max Delbrück Center · 2019 — 2026 (expected)

Thesis: Computational modelling and machine learning for spatial transcriptomics and cell–cell interactions.

M.Sc. in Computer Science

Technical University of Munich · 2016 — 2019

Thesis: Analysis of activity-dependent proteomic changes associated with homeostatic scaling. DAAD MSc Scholarship.

B.Sc. in Computer Science

Ozyegin University, Istanbul · 2012 — 2016

Thesis: A standalone application for a 2D ultrasound-guided medical robotic system.

SELECTED PUBLICATIONS

Ciltacabtagene autoleucel biomarkers of response in relapsed/refractory multiple myeloma patients from CARTITUDE-1: a phase 1b/2 study

R.M. de Oca, F. Wang, M. Hao, D. Smirnov, ... E. Senel, ... V. Plaks

Blood Cancer Discovery — Under revision · 2025

Optocoder: computational decoding of spatially indexed bead arrays

E. Senel, N. Rajewsky, N. Karaiskos

NAR Genomics & Bioinformatics · 2022

novoSpaRc: flexible spatial reconstruction of single-cell gene expression with optimal transport

N. Moriel*, E. Senel*, N. Friedman, N. Rajewsky, N. Karaiskos, M. Nitzan

Nature Protocols · 2021

Particulate impurities in cell-based medicinal products traced by flow imaging microscopy combined with deep learning for image analysis

A.D. Grabarek, E. Senel, T. Menzen, K.H. Hoogendoorn, K. Pike-Overet, A. Hawe, W. Jiskoot

Cytotherapy · 2021

Gabor filter-based localization of straight and curved needles in 2D ultrasound images

M. Kaya, E. Senel, O. Bebek

Turkish Journal of Electrical Engineering & Computer Sciences · 2020

Visual needle tip tracking in 2D US guided robotic interventions

M. Kaya, E. Senel, A. Ahmad, O. Bebek

Mechatronics · 2019

Visual tracking of multiple moving targets in 2D ultrasound guided robotic percutaneous interventions

M. Kaya, E. Senel, A. Ahmad, O. Bebek

ICRA, Singapore · 2017

Visual tracking of biopsy needles in 2D ultrasound images

M. Kaya, E. Senel, A. Ahmad, O. Bebek

ICRA, Stockholm · 2016

Real-time needle tip localization in 2D ultrasound images for robotic biopsies

A. Ahmad, O. Orhan, E. Senel, O. Bebek

ICAR, Istanbul · 2015

PATENTS

Method and system for 3D reconstruction of tissue gene expression data

S. Abbiati, J. Alles, S. Ayoub, S. Ehrig, N. Karaikos, G. Macino, S. Preibisch, N. Rajewsky, E. Senel, A. Woehler

US20240257914A1 · 2022

SELECTED OPEN-SOURCE PROJECTS

Optocoder — ML pipeline for decoding barcoded transcripts from imaging-based spatial transcriptomics.

github.com/rajewsky-lab/optocoder

novoSpaRc — Optimal-transport-based reconstruction of spatial gene expression from scRNA-seq.

github.com/rajewsky-lab/novosparc

SERVICE & COMMUNITY

Helmholtz Juniors — Elected Member (2022 — 2023): Represented ~8,000 PhD students across the Helmholtz Association; contributed to the 2021 PhD survey analysis and per-center reports.

MDC PhD Representatives — Awareness Team (2020 — 2023): Worked on contracts, salaries, and working conditions for doctoral students at the Max Delbrück Center.

LANGUAGES

English (Fluent) · German (Advanced) · Turkish (Native)